Questions

Q1.

A student is interested in the way that submarines are controlled.

She has several regular wooden blocks, a set of weights and a tank of water. Wood floats in water.

The student plans to immerse the wooden blocks fully in the water and investigate the relationship between the upthrust and the weight of water displaced.

Describe how she should determine **one** of the variables in this investigation.

(2)

.....

.....

.....

.....

(Total for question = 2 marks)

Q2.

Some questions must be answered with a cross in a box (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

A student carries out an investigation to show how pressure varies with depth in water.

A pressure sensor is attached to a rule.

The rule and pressure sensor are lowered into the water in a tank, as shown in Figure 8.

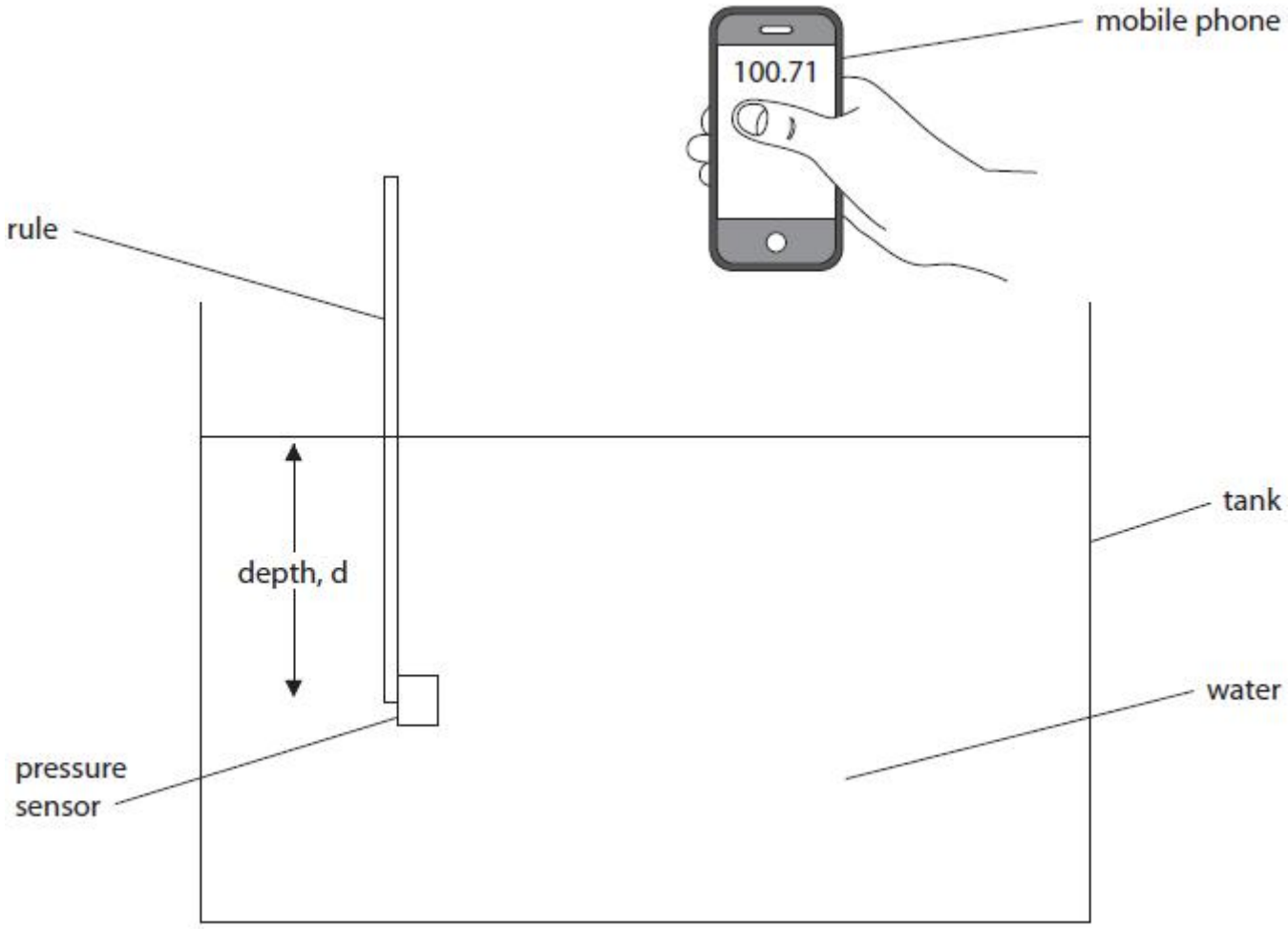


Figure 8

The depth of the pressure sensor below the surface of the water is read from the scale on the rule.

The pressure is displayed on the student's mobile phone which receives a signal from the pressure sensor.

Figure 9 gives some of the readings.

depth in m	pressure in kPa
0.050	99.15
0.100	99.70
0.150	100.15
0.200	100.70
0.250	101.15
0.300	101.70

Figure 9

Figure 10 shows a graph with some of the results plotted, but two of the points are missing.

Figure 10 shows a graph with some of the results plotted, but two of the points are missing.

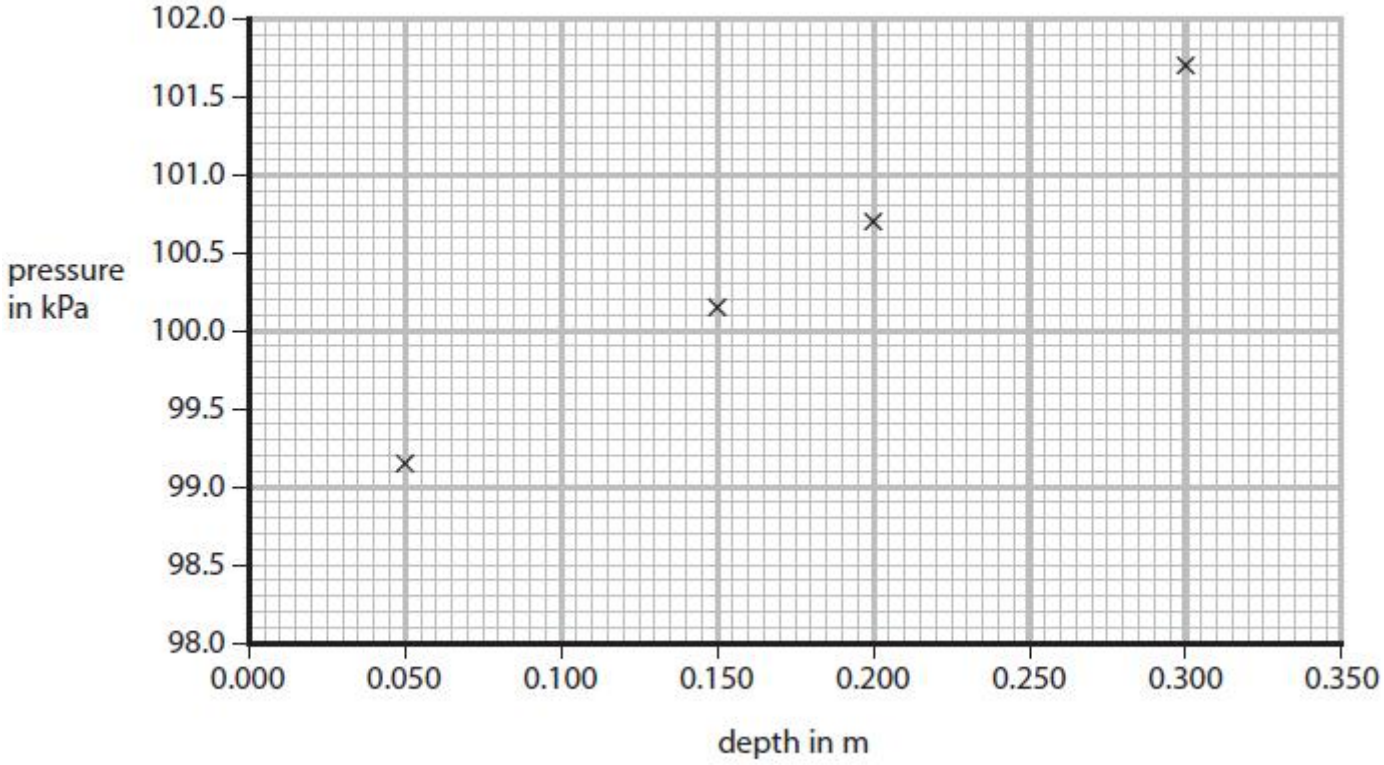


Figure 10

(i) Plot the two missing points on the graph.

(2)

(ii) Draw a line of best fit through the points on the graph.

(1)

(iii) Which of the following equations represents the variation of pressure with depth of water below the surface?

(1)

- ☐ **A** $y = ax^2 + b$
- ☐ **B** $y = mx$
- ☐ **C** $Y = mc - x$
- ☐ **D** $y = mx + c$

(iv) Use the graph in Figure 10 to predict the pressure at the surface of the water.

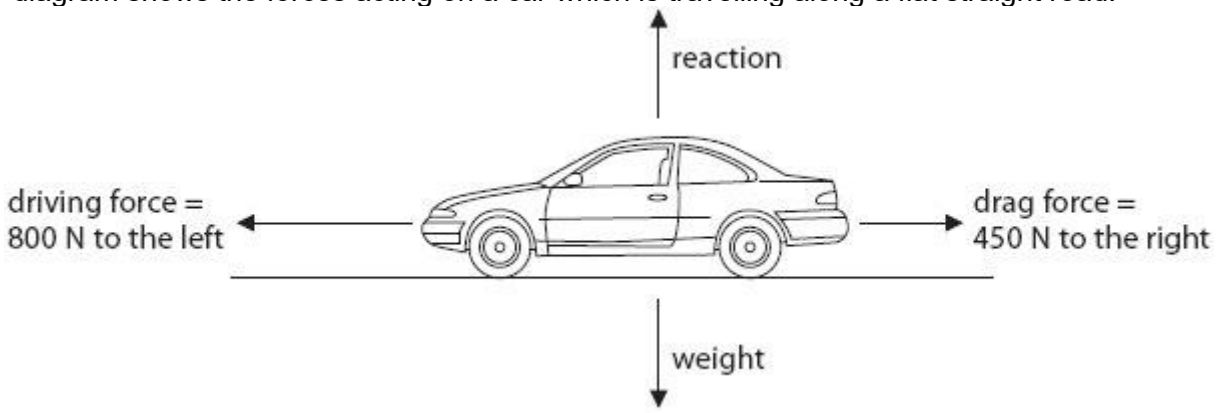
(1)

pressure at the surface of the sea water = ... kPa

(Total for question = 5 marks)

Q3.

The diagram shows the forces acting on a car which is travelling along a flat straight road.



(a) (i) The size of the resultant force on the car is 350 N.
In which direction is the resultant force acting?
Put a cross (■) in the box next to your answer.

(1)

- ☐ **A** down ↓
- ☐ **B** to the left ←
- ☐ **C** to the right →
- ☐ **D** up ↑

(ii) Complete the sentence by putting a cross (■) in the box next to your answer.

The car is

(1)

- ☐ **A** accelerating

☐ **B** decelerating

☐ **C** moving at a constant speed

☐ **D** not moving

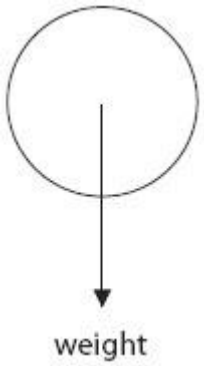
(2)

(iii) The mass of the car is 625 kg.
Calculate the weight of the car.
gravitational field strength = 10N/kg

(2)

.....

(b) Forces also act on objects when they fall through the air.
There are two forces acting on this ball as it falls through the air.
The weight is shown on the diagram.



(i) Draw and label an arrow on the diagram to show the other force acting on the ball.

(2)

(ii) Use words from the box to complete the sentences.

(2)

balanced	changing	greater	smaller	zero

After a short time the ball falls at a steady speed.

The forces acting on the ball are now .....

The acceleration of the ball is now .....

(Total for Question is 8 marks)

Q4.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

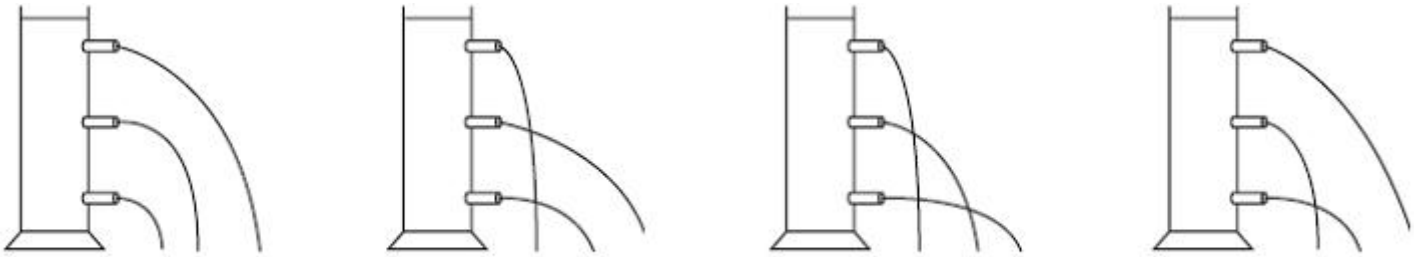
Figure 25 shows four identical metal cans, each filled with water to the same level.

Each can has three tubes.

Water comes out of each tube.

Which of these shows the correct pattern of water coming out of the tubes?

(1)



☐ **A** ☐ **B** ☐ **C** ☐ **D**

(Total for question = 1 mark)

Q5.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Figure 1 shows a pool full of water.

(Source: © AnaKulagina/Shutterstock)

Figure 1

A smaller pool has the same depth of water but has only half the area in contact with the ground.

The pressure on the ground underneath the smaller pool is

(1)

☐ **A** twice the pressure underneath the larger pool
☐ **B** ten times the pressure underneath the larger pool
☐ **C** half the pressure underneath the larger pool
☐ **D** the same as the pressure underneath the larger pool

(Total for question = 1 mark)

Q6.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Figure 18 shows a different spring hanging from a hook fixed to the ceiling.

A block hangs from the other end of the spring.

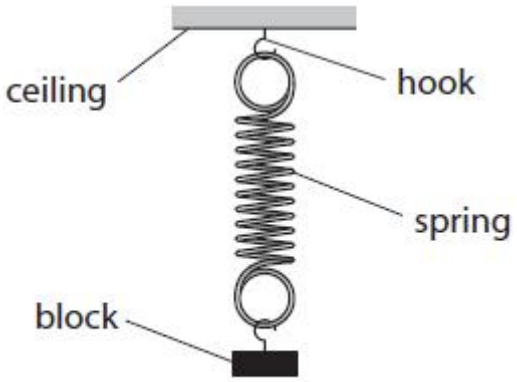


Figure 18

The weight of the spring is 1 N.
The weight of the block is 5 N.

The force exerted on the top of the spring by the hook is

(1)

☐ **A** 4 N down
☐ **B** 4 N up
☐ **C** 6 N down
☐ **D** 6 N up

(Total for question = 1 mark)

Q7.

Figure 2 shows a tank for holding water.

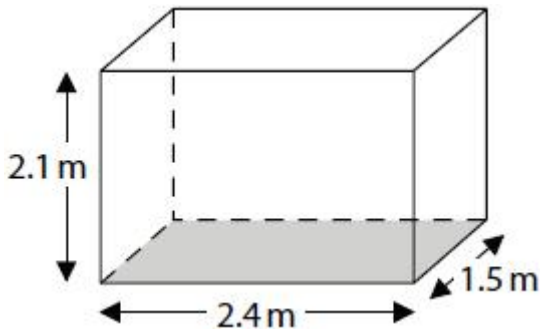
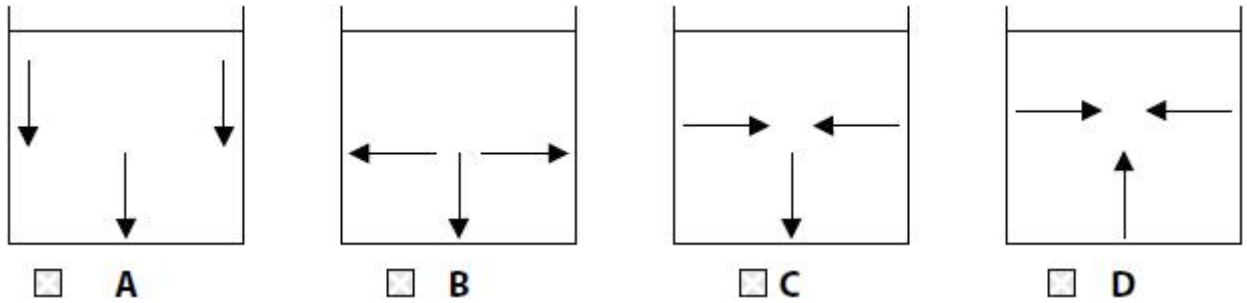


Figure 2

Which diagram shows the direction of the forces from the water on the inside of the tank?

(1)



(Total for question = 1 mark)

Q8.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Figure 9 shows an object at the bottom of a beaker of water.

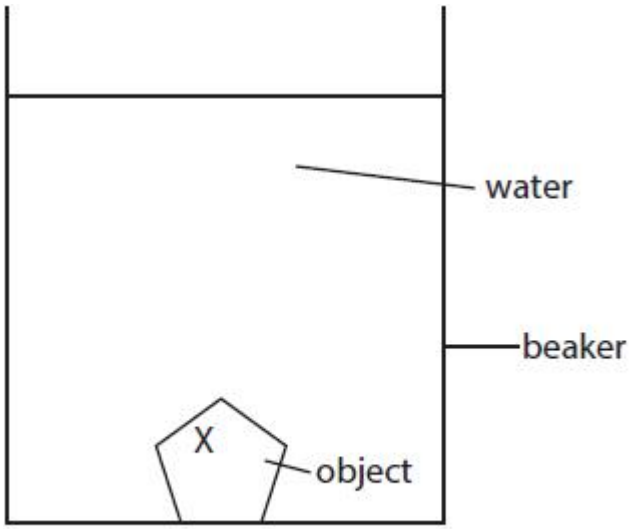
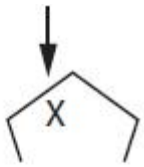


Figure 9

Which diagram shows the direction of the force exerted by the water on the object at point X?

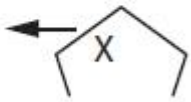
(1)

☐ **A**

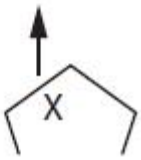


☐ **B**

☐ **C**



☐ **D**



(Total for question = 1 mark)

Q9.

Figure 10 is a graph showing how the atmospheric pressure changes with the height above sea level on the Earth's surface.

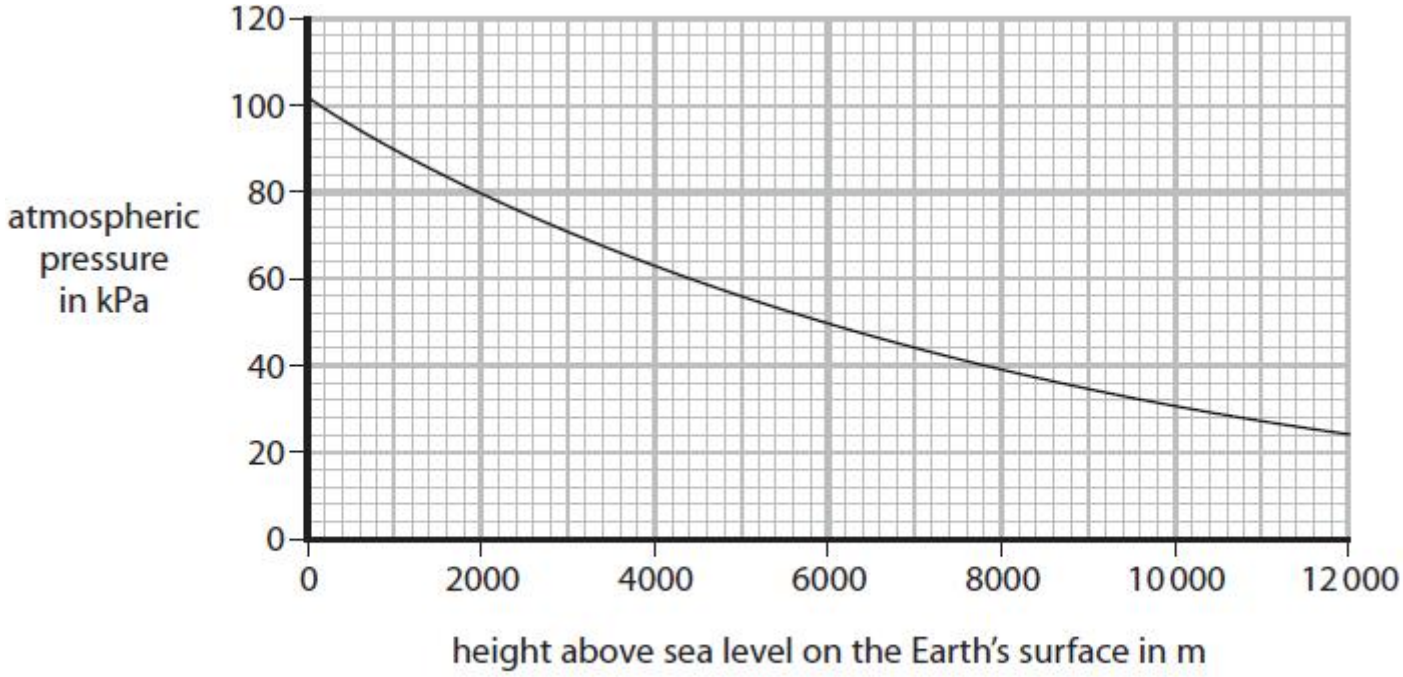


Figure 10

(i) An aeroplane descends from 6000 m to 2000 m.

Use the graph to find the change in atmospheric pressure as the aeroplane descends.

(2)

change in pressure = ... kPa

(ii) Suggest **one** reason why the atmospheric pressure is greater at 2000 m than at 6000 m.

(1)

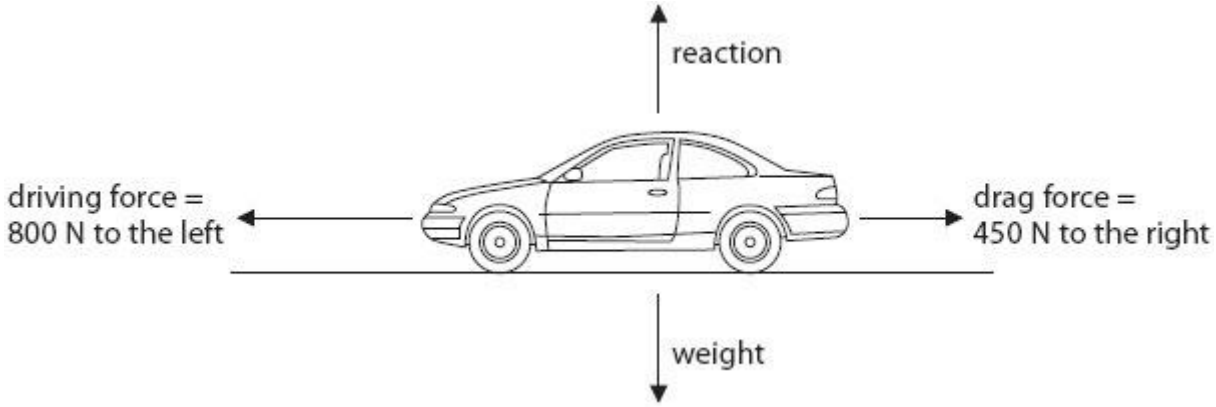
.....

.....

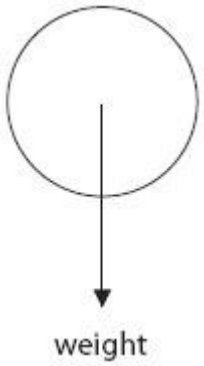
(Total for question = 3 marks)

Q10.

The diagram shows the forces acting on a car which is travelling along a flat straight road.



Forces also act on objects when they fall through the air.
There are two forces acting on this ball as it falls through the air.
The weight is shown on the diagram.



(i) Draw and label an arrow on the diagram to show the other force acting on the ball.

(2)

(ii) Use words from the box to complete the sentences.

(2)

balanced	changing	greater	smaller	zero

After a short time the ball falls at a steady speed.

The forces acting on the ball are now .....

The acceleration of the ball is now .....

Q11.

Figure 13 shows a diagram of a device for lifting heavy loads.

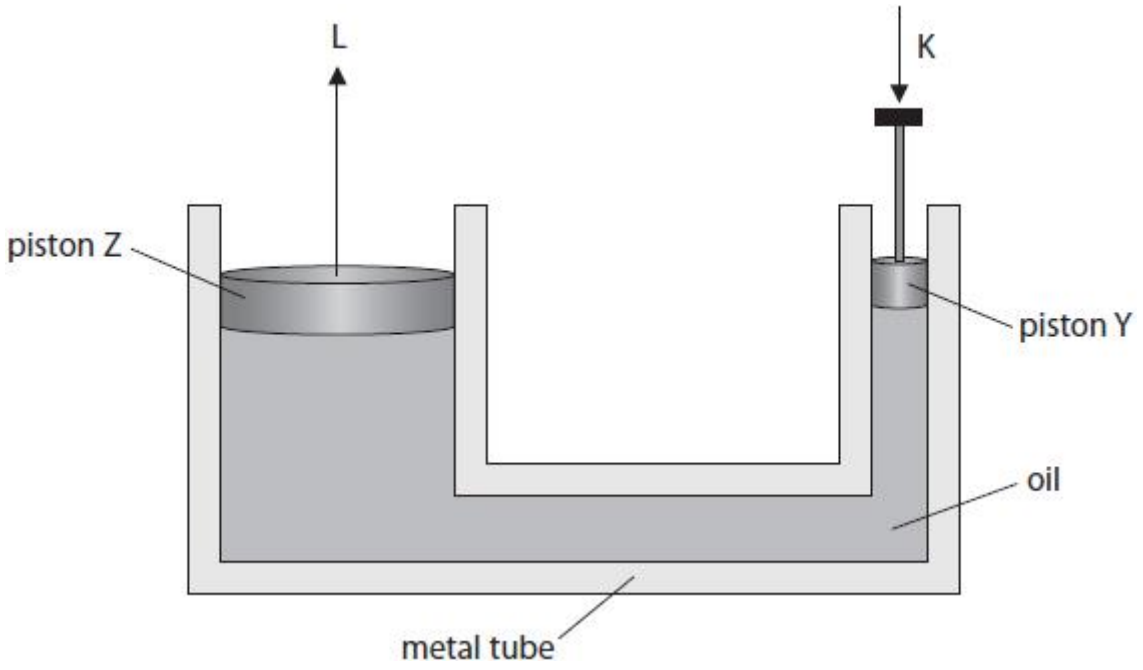


Figure 13

The metal tube is filled with oil.

The piston Y is pushed down with a force K.

This produces a force L on piston Z.

The pressure exerted on the oil by piston Y is the same as the pressure exerted by the oil on piston Z.

Explain the difference between the size of force K and the size of force L.

(3)

.....

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.....

.....

.....

(Total for question = 3 marks)

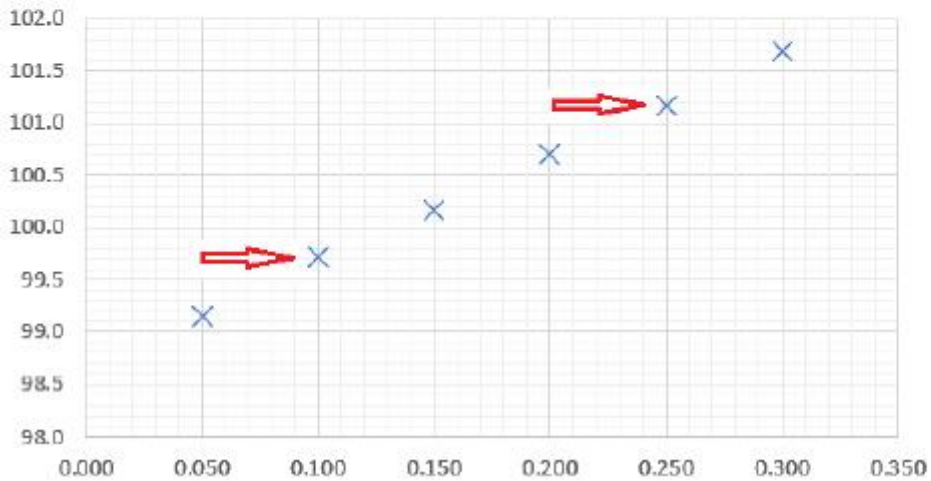
Mark Scheme

Q1.

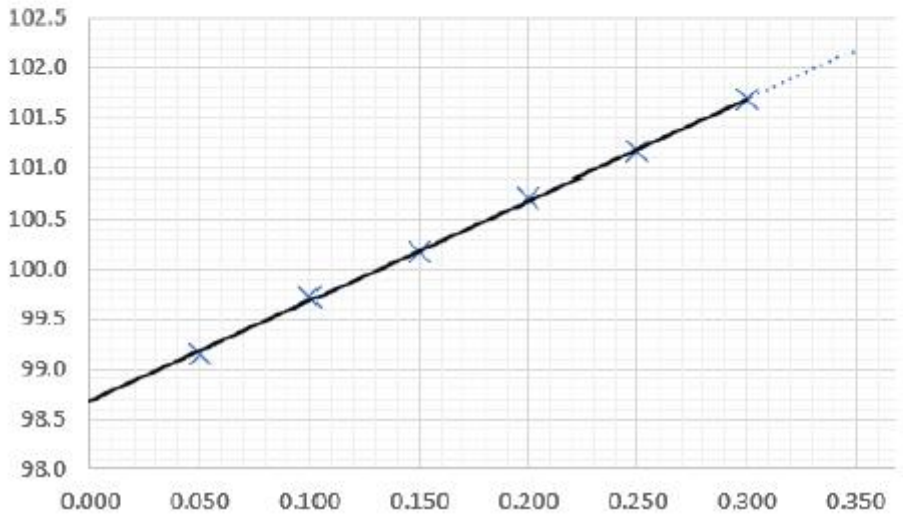
Question number	Answer	Mark
	An answer that combines the following points to provide a logical description of the plan: EITHER • (determine upthrust) by adding weights until the block of wood is fully immersed and recording the load required (1) • calculate upthrust by adding load and weight of block (1) OR • (determine the weight of water displaced) by using a ruler to measure the dimensions of the block and multiplying them together to find the volume (1) • calculate the weight of water from volume × density × g (1)	(2)

Q2.

Question number	Answer	Additional guidance	Mark
(i)	points plotted to within ± 1 small square (0.100, 99.7) (1) (0.250, 101.15) (1)		(2) AO2.1



Question number	Answer	Additional guidance	Mark
(ii)	best fit straight line passing through at least four of the points (1)	do not accept tramlining (multiple lines / curves) ignore slight shakiness in drawing	(1) AO2.1



Question number	Answer	Mark
(iii)	D $y = mx + c$ Figure 10 shows a linear graph with a positive gradient and intercept	(1) AO1.1

Question number	Answer	Additional guidance	Mark
(iv)	answer between 98.6 and 98.8 (kPa)	allow ecf from their line of best fit in b(ii)	(1) AO3.2

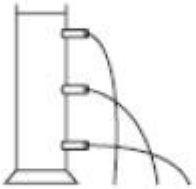
Q3.

	Answer	Acceptable answers	Mark
(a)(i)	B to the left ←		(1)
(a)(ii)	A accelerating		(1)
(a)(iii)	substitution 625x 10 (1) Evaluation 6250 (N)	625 x 9.8 6125 (N) give full marks for correct answer, no working	(2)

	(1)		
(b)(i)	(1) air resistance (1)	upward arrow on any part of line vertical line from any point on the diagram air friction, upthrust, drag Ignore any downward arrow labelled weight or gravity	**(2)**
(b)(ii)	Balanced (1) Zero (1)		**(2)**

Total for marks for question = 8

Q4.

Question number	Answer	Mark
	C  A,B and D are incorrect because pressure increases with depth	**(1)**

Q5.

Question number	Answer	Additional guidance	Mark
	D the same as the pressure underneath the large pool A, B and C are all incorrect because the pressure is the same		**(1)** **AO2.2**

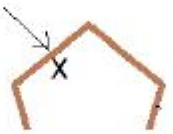
Q6.

Question number	Answer	Additional guidance	Mark
	D 6 N up A and C are incorrect because the force is upwards B is incorrect because the force is the sum of the two weights given.		**(1)** **AO3**

Q7.

Question number	Answer	Mark
	B	**(1)**

Q8.

Question number	Answer	Mark
	B  This shows the **only** direction normal to surface, acting towards surface	**1** **AO3.1**

Q9.

Question number	Answer	Additional guidance	Mark
i	data points correctly identified (1) 50 ±2 80 ±2 evaluation (1) (-)30 ±4 kPa	award 1 mark if 80 and 50 seen ignore the lack of minus sign allow ecf from incorrect reading of either pressure at 2000m or pressure at 6000m for one mark	**(2)** **AO3**

Question number	Answer	Additional guidance	Mark
	any **one** suggestion of greater density of atmosphere (1) greater depth of atmosphere (above the aeroplane) (1) greater temperature (of the atmosphere) (1)	accept reverse argument more particles (per cubic metre) the air gets thicker greater weight of the atmosphere	**(1)** **AO1**

Q10.

	Answer	Acceptable answers	Mark
(i)	 (1) air resistance (1)	upward arrow on any part of line vertical line from any point on the diagram air friction, upthrust, drag Ignore any downward arrow labelled weight or gravity	**(2)**
(ii)	Balanced (1) Zero (1)		**(2)**

Q11.

Question Number:	Answer	Additional Guidance	Mark
	an explanation linking: use of $P = \frac{F}{A}$ (1) **area** of piston Y is less than area of piston Z (1) (therefore) force K is less than force L (1)	accept answers in terms of work = force x distance accept reverse arguments accept K for piston Y and L for piston Z	**(3)** AO 3 2a AO 3 2b